

Simplification of Context free grammar:

Defn -

In CFG, Sometimes all the production rules & Symbols are not needed for the derivation of strings.

Besides this, there may also be some NULL Productions & UNIT productions. Elimination of these productions & Symbols is called simplification of CFG.

Simplification consists of the following steps.

- ① Reduction of CFG. 3
- ② Removal of unit production. 2
- ③ Removal of NULL production. 1 ↑ first remove.

① Reduction of CFG

phase-I

phase-II

Phase-I → Derivation of an equivalent grammar G' from the CFG G , such that each variable derives some terminal string.

Derivation Procedure

- Step 1: Include all symbols w_i , that derives some terminal & initialize $i=1$.
- Step 2: Include symbol w_{i+1} , that derives w_i .
- Step 3: Increment i & repeat Step 2, until $w_{i+1} = w$.
- Step 4: Include all production rules that have w_i in it.

$B \rightarrow Bala$

not from (GMP)
be in GMP

for

$A \rightarrow a$

non-terminal

Phase 2 Derivation of an equivalent grammar G' ,
from the CFG, G , such that each symbol
appears in a sentential form.

Derivation Procedure:

- Step 1: Include the start symbol in Y_1 & initialize $i=1$
- Step 2: Include all symbol Y_{i+1} , that can be
derived from Y_i & include all production rules
that have been applied.
- Step 3: Increment i & repeat step 2, until $Y_{i+1} = Y_i$

$B \rightarrow 15a1a$

3 Find the reduced grammar equivalent to the grammar G , having production rules as

$$S \rightarrow AC|B$$

$$A \rightarrow a$$

$$C \rightarrow c|BC$$

$$E \rightarrow QA|e$$

Soln

Phase I.

Terminal symbol are

$$T = \{a, c, e\}$$

All the ^{Non-}terminal that derive T

$$w_1 = \{A, C, E\}$$

Include ^{in w_1} All the symbol that can be derive from w_1

$$w_2 = \{A, C, E, S\}$$

Include all the symbol that can derive w_2

$$w_2 = \{A, C, E, S\}$$

$w_2 = w_3$ so stop.

B is not here. So remove it.

Our grammar $G' = (\{A, C, E, S\}, \{a, c, e\}, P, S)$ become

production P : $S \rightarrow AC, A \rightarrow a, C \rightarrow c, E \rightarrow QA|e$

Phase-II

Look at G'

Include the start symbol.

$$Y_1 = \{S\}$$

All the symbol that can be derive from start S .

$$Y_2 = \{S, A, C\}$$

All the symbol that can be derive from Y_2

$$Y_2 = \{S, A, C, a, c\} = \{S, A, C\} \cup \{a, c\}$$

Capital
Small

$B \rightarrow Ba|a$

(GMP)

is the

All symbols that can be derived from y_3

$$y_4 = \{s, A, C, a, c\}$$

~~$y_3 = y_4$~~ Step

$$y_2 = y_4$$

$$G'' = \{ (A, C, s) (a, c), P, S \}$$

Production P

$s \rightarrow AC$
$A \rightarrow a$
$C \rightarrow c$

G''
Reduce
grammar.

$\rightarrow$

Removal of unit Production

Any production rule of the form $A \rightarrow B$ where $A, B \in \text{Non Terminals}$ is called unit production.

Procedure:

Step 1 $\rightarrow$ To remove $A \rightarrow B$, add $A \rightarrow x$ to the grammar rule whenever $B \rightarrow x$ occurs in the grammar ($x \in \text{terminal}$, x can be Null.)

Step 2 - delete $A \rightarrow B$ from the grammar.

Step 3 $\rightarrow$ Repeat from Step 1 until all unit production are removed.

$B \rightarrow Ba|a$ (GMF) $\therefore$ $\therefore$
Eliminate the useless production from the given grammar. Enlist all step clearly.

$$S \rightarrow aAB|aBA|aAS$$

$$A \rightarrow aA$$

$$B \rightarrow ab|b$$

$$C \rightarrow ad.$$

Terminal are (a, b, ad)

Solⁿ Phase-I

Step 1 $\rightarrow$ Find generating symbol using $w_1, w_2, \dots$
i.e. Variable that only produce terminals.

$$B \rightarrow ab|b$$

$$C \rightarrow ad.$$

$$\text{So } w_1 = \{B, C\}.$$

Now check Variables that can produce terminals using Nonterminal of w_1 .

i.e. $S \rightarrow aAB$ i.e. $B \in w_1$ & it generating terminal.
 $\hookrightarrow A \rightarrow aA$ i.e. $A \notin w_1$.

$$\text{So } w_2 = \{B, C, S\}.$$

Here No other Variable is available which determine terminal symbol so

$$w_3 = w_2 \text{ & stop.}$$

So the $G' = (S, B, C, \{a, b, ad\}, P, S)$.
Grammar become $S \rightarrow aAB|aAS$

$$B \rightarrow ab|b$$

$$C \rightarrow ad.$$

Phase-II Look at G' find start symbol.

$$Y_1 = \{S\}$$

$$B \rightarrow Ba | a$$

HF)

$$\text{from } S \rightarrow aab | aaS$$

B is reachable from start.

S is already included.

$$\text{So } Y_2 = \{S, B\}$$

there is no any other variable reachable from Y_2

$$\text{So } Y_2 = Y_3 \text{ . stop .}$$

$$G'' = \{(S, B) (ab, b), P, S\}$$

So the reduce grammar is.

$$\begin{array}{l} S \rightarrow aab | aaS \\ B \rightarrow ab | b \end{array}$$

Final Ans

$$w_1 = \{B, c\} \checkmark$$

$$w_2 = \{S, B, c\} \checkmark$$

$$\text{Generating} = \{S, B, c\} \checkmark$$

$$\text{Non generating} = \{A\} \times$$

$$V_1 = \{S\} \checkmark$$

$$V_2 = \{S, B\} \checkmark$$

$$\text{Reachable} = \{S, B\} \checkmark$$

$$\text{Unreachable} = \{c\} \times$$

eliminate the useless production from the given grammar. (Repeat)

$S \rightarrow aabB | abaA | aas$

$A \rightarrow aA$

$B \rightarrow ab|b$

$C \rightarrow ad$

Solⁿ phase I $\rightarrow$

Terminal symbol are

$T = \{ab, b, ad\}$

All the Non Terminal that derive T .

$w_1 = \{B, C\}$

All the symbol that derive w_1 is

$w_2 = \{S, B, C\}$

All the symbol that derive w_2

$w_3 = \{S, B, C\}$

Here $w_2 = w_3$ stop.

Apr grammar

$G' = (\{S, B, C\}, \{ab, b, ad\}, P, S)$

$S \rightarrow aabB | aas$

$B \rightarrow b|ab$

$C \rightarrow ad$

Phase-II Look at G'

Include start

$y_1 = \{S\}$

All the symbol that derive from S

$y_2 = \{aabB, aas\}$

All the symbol that derive from y_2

$y_3 = \{S, aabB, aas, ab, b\}$

symbol that can be derive from y_3

$y_4 = \{S, aabB, aas, ab, b\}$

$y_3 = y_4$ stop.

$G'' = (\{S, B\}, \{ab, b\}, P, S)$

$S \rightarrow aabB | aas$
 $B \rightarrow ab|b$ Reduce grammar.

Eliminate the useless productions from the given grammar.

$$S \rightarrow AB \mid AC$$

$$A \rightarrow aAb \mid bAa \mid a$$

$$B \rightarrow bBA \mid aAB \mid AB$$

$$C \rightarrow abCa \mid aDb$$

$$D \rightarrow bD \mid aC.$$

Soln. Phase - I

Variable that directly produce only terminal

$$T = \{a\} \quad \therefore A \rightarrow a.$$

$$\text{So } W_1 = \{A\}$$

Add variables that can produce # string using only W_1 .

$$W_2 = B \rightarrow bBA \quad \text{as } A \in W_1 \rightarrow \text{include } B.$$

$$S \rightarrow AB \quad \text{as } A, B \in W_1 \cup W_2 \rightarrow \text{include } S$$

$$\begin{array}{l} C \rightarrow abCa - X \\ C \rightarrow aDb - X \end{array} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} \text{Not include because} \\ \text{they are not from } A \text{ or } B \end{array}$$

$$\text{So, } W_2 = \{A, B, S\}.$$

Include all the symbols that can derive from W_2

$$W_2 = W_2 = \{A, B, S\}.$$

$$\text{So } G' = (\{A, B, S\}, \{a\}, P, S).$$

$$\begin{array}{l} \cancel{S \rightarrow AB \mid AC} \quad S \rightarrow AB \\ A \rightarrow aAb \mid bAa \mid a \\ B \rightarrow bBA \mid aAB \mid AB. \end{array}$$

Phase II

Look at G' - Include the $\#$ start symbol.

$$V_1 = \{S\}$$

All the symbols that derive from S .

$$V_2 = \{A, B\}$$

All the symbols that derive from V_2

$$V_3 = \{A, B, a, aAb, bAa, bBA, aAB, AB\}.$$

All the remaining symbols that derive from y_3

$$y_4 = \{A, B, a, aAb, bAa, bBa, bAb, aAB, AB\} -$$

$$= y_3$$

Stop.

$$G' = \{ \{A, B, S\}, \{a, b\}, P, S \}$$

So

$$\begin{cases} S \rightarrow AB \\ A \rightarrow aAb \mid bAa \mid a \\ B \rightarrow bBa \mid aAB \mid AB \end{cases}$$

Reduce Grammar.

Ans (1)

$$S \rightarrow AB \mid Ba$$

$$A \rightarrow a$$

$$B \rightarrow C$$

$$C \rightarrow bC \mid \epsilon$$

Simplify the above
~~Context~~ Context free
grammar.

Ans (2)

$$S \rightarrow ABC \mid BaB$$

$$A \rightarrow aA \mid BaC \mid aaa$$

$$B \rightarrow bBb \mid a$$

$$C \rightarrow CA \mid AC$$

Convert the following
grammar into an
equivalent one with
no unit productions
and no useless symbols

$$\Sigma \rightarrow \{aaa, a, b\}$$

$$w_0 \rightarrow A, B,$$

$$w_1 \rightarrow S, AB$$

$$G = \{ \{A, B, S\}, \{aaa, a, b\}, P, S \}$$

$$S \rightarrow ABaB$$

$$A \rightarrow aA \mid aaa$$

$$B \rightarrow bBb \mid a$$

$$y_1 = \{S\}$$

$$y_2 = \{S, BaB\}$$

$$y_3 = \{S, BaB\}$$

$$G'' = \{ \{S, B\}, \{aaa, a, b\} \}$$

$$\begin{cases} S \rightarrow BaB \\ B \rightarrow bBb \mid a \end{cases}$$

Q. Eliminate useless grammar.

$$S \rightarrow AB/AC$$

$$A \rightarrow aAb/bAa/a$$

$$B \rightarrow bBA/aAB/AB$$

$$C \rightarrow abCa/aDb$$

$$D \rightarrow bD/aC$$

Soln
Phase 1

$$T = \{a\}$$

$$w_1 = \{A\}$$

$$w_2 = \{S, A, B\}$$

$$w_3 = w_2 \cap \{S, A, B\}$$

= w_2 stop

$$G' = (\{S, A, B\}, \{a\}, P, S)$$

$$S \rightarrow AB$$

$$A \rightarrow aAb/bAa/a$$

$$B \rightarrow bBA/aAB/AB$$

Phase 2

$$\text{Step 1 } V_1 = \{S\}$$

$$V_2 = \{A, B, S\}$$

$$V_3 = \{A, B, aAb, bAa, a, bBA, aAB, AB\}$$

$$V_4 = V_3 \text{ stop}$$

$$G'' = (\{S, A, B\}, \{a, b\}, P, S)$$

$$S \rightarrow AB$$

$$A \rightarrow aAb/bAa/a$$

$$B \rightarrow bBA/aAB/AB$$

Convert the below given PDA into CFG

$$\delta(q_0, a, z_0) = (q_0, xz_0)$$

$$\delta(q_0, a, x) = (q_0, xx)$$

$$\delta(q_0, b, x) = (q_1, \epsilon)$$

$$S \rightarrow AB | Ba$$

$$A \rightarrow a$$

$$B \rightarrow \epsilon$$

$$C \rightarrow bC | \epsilon$$

Simplify the grammar.

Soln

Epsilon Remove

$$C \rightarrow \epsilon \text{ so.}$$

$$w_0 = \{C\}$$

Indirectly

$$w_1 = \{C, B\}$$

So our grammar become.

Expand.

$$S \rightarrow AB | Ba$$

$$A \rightarrow a$$

$$B \rightarrow \epsilon | C$$

$$C \rightarrow bC | b$$

$$S \rightarrow AB | Ba | A | a$$

$$A \rightarrow a$$

$$B \rightarrow C$$

$$C \rightarrow bC | b$$

New unit production $S \rightarrow B$ Remove it

$$\text{for } S \quad w_0(S) = \{S, A\}$$

$$w_1(S) = w_0(S) \cup \phi$$

$$= \{S, A\} = w_0(S)$$

for B

$$w_0(B) = \{B, C\}$$

$$w_1(B) = w_0(B) \cup \phi$$

$$\Rightarrow w_0(B)$$

$$\Rightarrow \text{stop}$$

$$\boxed{S \rightarrow AB | Ba | a}$$

$$A \rightarrow a$$

$$C \rightarrow bC | b$$

New eliminate useless production.

Phase-I

$$T = \{b, a\}$$

$$w_1 = \{S, C, A\} \quad \{S, C, A\}$$

$$w_2 = \{S, A\} = w_1 = \{S, C, A\}$$

$$G' = \{(S, A), (a, b), (p, S)\}$$

$$S \rightarrow a, A \rightarrow a$$

$$C \rightarrow b | bC$$

Phase 2

Start

$$V_1 = \{S\}$$

$$V_2 = \{S, a\}$$

$$V_3 = \{S, a\}$$

$$G'' = \{S, a, (p, S)\}$$

So

$$\boxed{S \rightarrow a}$$

Reduce grammar.